Physics 710-711-712 November 25, 2009

Problem Set 6

Problem 1: Exercise 5.1.2

In the X basis, (5.1.3) becomes $\frac{-\hbar^2}{2m}\psi''_E = E\psi_E$, whose solutions are $\psi = Ae^{ikx} + Be^{-ikx}$ where $k\hbar = \sqrt{2mE}$. This is the solution given in the exercise. If $E < 0$, these solutions are not in the Hilbert space, since then the two terms grow exponentially either as $x \to +\infty$ or as $x \to -\infty$.

Problem 2: Exercise 5.2.2

(a) Say $\{|n\rangle\}$ is the orthonormal basis of H eigenstates, so $H|n\rangle = E_n|n\rangle$ and $1 = \sum_n |n\rangle\langle n|$. Then

$$\begin{aligned}
\langle\psi|H|\psi\rangle &= \sum_n \langle\psi|H|n\rangle\langle n|\psi\rangle = \sum_n E_n\langle\psi|n\rangle\langle n|\psi\rangle = \sum_n E_n|\langle n|\psi\rangle|^2 \\
&\geq \sum_n E_0|\langle n|\psi\rangle|^2 = E_0\sum_n \langle\psi|n\rangle\langle n|\psi\rangle = E_0\langle\psi|\psi\rangle = E_0.
\end{aligned}$$

The inequality in the second line follows because, by definition, $E_0 < E_n$ for all $n > 0$.

(b)

$$\begin{aligned}
E(\alpha) &:= \langle\psi_\alpha|H|\psi_\alpha\rangle = \int_{-\infty}^{\infty} dx\, \psi^*_\alpha(x)\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} - |V(x)|\right)\psi_\alpha(x) \\
&= \sqrt{\frac{\alpha}{\pi}}\int_{-\infty}^{\infty} dx\, e^{-\alpha x^2/2}\left(-\frac{\hbar^2}{2m}\left[-\alpha + \alpha^2 x^2\right] - |V(x)|\right)e^{-\alpha x^2/2} \\
&= \sqrt{\frac{\alpha}{\pi}}\left(\frac{\hbar^2}{2m}\frac{\sqrt{\alpha\pi}}{2} - \int_{-\infty}^{\infty} dx\, |V(x)|e^{-\alpha x^2}\right).
\end{aligned}$$

By taking $\alpha \to 0$, you can make the first (positive) term inside the parentheses as small as you like, while the second (negative) term appraoches the finite negative value $-\int dx\, |V|$. Thus we can make $E(\alpha) < 0$, and therefore, by part (a), $E_0 < 0$.

Problem 3: Exercise 5.2.6

(1) Call $x < -a$ region I, $-a < x < a$ region II, and $x > a$ region III. Then solving for the energy eigenstates, $\hbar\psi'' = 2m(E - V(x))\psi$, of energy $E \leq V_0$ in each region gives $\psi_I = Ae^{-\kappa x} + Be^{\kappa x}$, $\psi_{II} = Ce^{ikx} + De^{-ikx}$, and $\psi_{III} = Ee^{-\kappa x} + Fe^{\kappa x}$, with $\hbar\kappa = \sqrt{2m(V_0 - E)}$ and $\hbar k = \sqrt{2mE}$. We must have $A = F = 0$ so that the wave function does not grow exponentially as $x \to \pm\infty$. The boundary conditions at $x = \pm a$ are that ψ and ψ' are continuous, implying

$$\begin{aligned} Be^{-\kappa a} &= Ce^{-ika} + De^{ika}, \\ \kappa Be^{-\kappa a} &= ikCe^{-ika} - ikDe^{ika}, \\ Ee^{-\kappa a} &= Ce^{ika} + De^{-ika}, \\ -\kappa Ee^{-\kappa a} &= ikCe^{ika} - ikDe^{-ika}, \end{aligned}$$

Taking the suggestion of the problem, let's look for even and odd solutions, that is, those satisfying $\psi(-x) = \psi(x)$ and $\psi(-x) = -\psi(x)$, respectively.
Even solutions: For ψ to be even, we need $B = E$ and $C = D$. The above boundary conditions then become

$$\begin{aligned} Be^{-\kappa a} &= Ce^{-ika} + Ce^{ika}, \\ \kappa Be^{-\kappa a} &= ikCe^{-ika} - ikCe^{ika}, \end{aligned}$$

which can be rewritten as

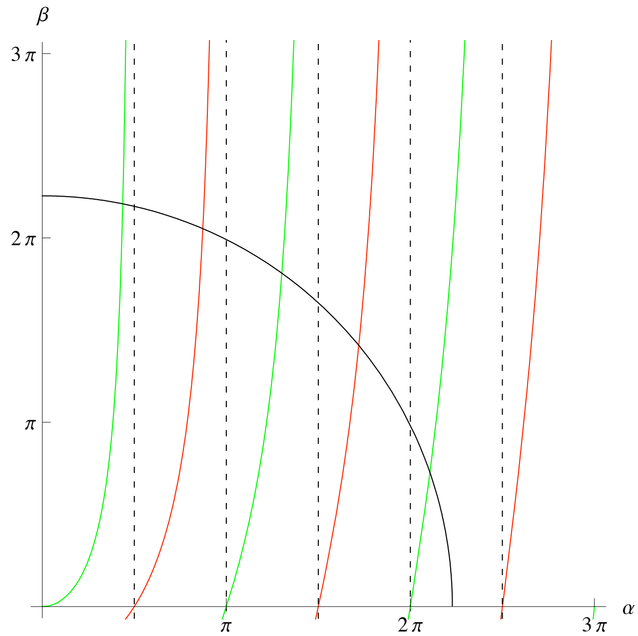
$$\begin{pmatrix} e^{-\kappa a} & -2\cos(ka) \\ \kappa e^{-\kappa a} & -2k\sin(ka) \end{pmatrix} \begin{pmatrix} B \\ C \end{pmatrix} = 0,$$

which has a solution for nonzero $(B\ C)$ only if the determinant of the matrix vansihes, implying $k\sin(ka) = \kappa\cos(ka)$, which is the result we wanted.
Odd solutions: For ψ to be even, we need $B = -E$ and $C = -D$. The above boundary conditions then become just two independent equations which can be rewritten as

$$\begin{pmatrix} e^{-\kappa a} & 2i\sin(ka) \\ \kappa e^{-\kappa a} & -2ik\cos(ka) \end{pmatrix} \begin{pmatrix} B \\ C \end{pmatrix} = 0.$$

The determinant of the matrix vansihes when $-k\cos(ka) = \kappa\sin(ka)$, which is the result we wanted.

(2) This part of the problem is just a plot. In the variables α and β defined in the problem, (5.2.25) becomes the curve of the circle $\alpha^2 + \beta^2 = 2ma^2V_0/\hbar^2$. Plotting the circle and the $\beta = \alpha\tan\alpha$ (even states) or $\beta = -\alpha\cot\alpha$ (odd states) then gives the plot shown below. The red (darker) curves are the odd states and the green (lighter) ones are the even ones, and the quarter-circle is plotted for the value $2ma^2V_0^2/\hbar^2 = 49$. (The dotted lines just indicate the asymptotes to the red and green curves.) Each intersection of the circle with a red or green line is a solution, and therefore an energy eigenstate. For the value plotted in the figure, we thus see that there are five eigenstates.



(3) As $V_0 \to 0$, the radius of the circle decreases. When it gets less than $\pi/2$ it can only intersect the leftmost green curve. Since that curve goes to the origin, no matter how small V_0, the circle will always intersect it, and so there will always be at least one even solution.

When the radius equals $\pi/2$, then $V_0 = \hbar^2\pi^2/8ma^2$, and the circle intersects the $\alpha\tan(\alpha)$ curve at $\alpha \approx 0.934014$ (found numerically). Since $\alpha := ka$ and $\hbar k = \sqrt{2mE}$, this implies $E = \hbar^2\alpha^2/2ma^2 \approx 0.46191\,\hbar^2/ma^2$.